Response to Reviewer 2

Manuscript: *Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams* (submitted as *Quantum Oracle Sketching for Non-IID Sensor and Telemetry Streams: A Proxy-Based Computational Study*)
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We thank both reviewers for careful, constructive reviews that have materially improved the paper. Both reviews converge on the same central point, and we agree with it: the previous version, although internally hedged, still presented itself as evidence about quantum performance, when what it can honestly support is a **proxy-based diagnostic and hypothesis-generation framework for correlation-sensitive streaming analysis**. The revision repositions the manuscript accordingly — title, abstract, statistics, figures, and conclusion — without removing any experiment or altering any measured result: all stochastic outputs regenerate bit-identically from the same seeded pipeline (we verified this explicitly before re-analysis; the verification scripts ship with the revision).

Summary of the principal changes (full detail under each point below):

1. **Retitled** to *Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams*; the abstract now opens "In theory, …", states that no quantum algorithm is implemented or simulated, and was shortened to ~220 words. *(Note on the tracked-changes file: the MDPI class places the title and abstract in the document preamble, which latexdiff does not mark; both are completely rewritten.)*
2. **New Section 3.3, "Relationship Between Operational Proxies and Quantum Oracle-Sketching Theory"**, separating established theory / assumptions / heuristic constructions / scope limitations, with a provenance table (Table 1) stating per quantity what is inherited, what is introduced, and what is *not* theoretically guaranteed — including the explicit statement that under the framework's own r-only accounting no proxy–baseline crossover exists, so the τ-penalised landscape is a classical effective-sample stress test, not the framework's cost model.
3. **Statistics rebuilt around estimation rather than testing** (Experiment 5): Hodges–Lehmann paired-difference estimates with exact signed-rank 95% intervals, matched-pairs rank-biserial correlations (interpreted as sign-concordance), p-values demoted to descriptive companions, disclosure that no comparison survives Holm correction at n = 10 (with the reason), the τ-estimator lattice sensitivity, and an explicit statement that the crossover location is a deterministic solution whose uncertainty comes from the τ estimator and the constants, not from tests.
4. **New sensitivity analyses from existing data** (no new stochastic runs): a C–δ grid for the crossover (Table 8: rho* moves from 0.38 to beyond the sweep; ordinal reading is the robust one) and an encoder/resolution sensitivity table for the real streams (Table: 10 re-encodings; the NYC-above-Machine-Temperature ordering holds in all 10).
5. **New Discussion subsections** on why analytic and empirical correlation estimates diverge, and on what the landscape can and cannot predict — including three **falsifiable hypotheses (H1–H3) with explicit refutation criteria**.
6. **New "Threats to Validity" section** (construct / internal / external / statistical-conclusion) and a promoted, expanded **Future Work** section.
7. **Conclusion rewritten**: primary contribution = the (τ, r) landscape; secondary = proxy-based hypothesis generation; the sentence "The paper demonstrates no quantum advantage and implements no quantum algorithm" appears verbatim.
8. **Figures**: the Experiment-5 legend entry "Proxy-model advantage" renamed "Proxy-estimated separation"; in-image labels "n = 10 qubits" / "Quantum O(n)" removed from Figures 1, 2, 6; captions

aligned with the proxy framing; Figure 8(d)'s synthetic markers disclosed as schematic; typographic and overfull fixes throughout.

9. **Reference audit (preprints and metadata)**: all 54 references were classified, and every entry was then verified at the metadata level — all 51 DOIs were resolved through doi.org content negotiation and machine-compared against the cited title, first author, venue, and year, with the non-DOI entries checked at source. Two corrections resulted: (i) Kallaugher, Parekh and Voronova (SOSA 2025) now carries the SIAM proceedings DOI (10.1137/1.9781611978315.2) instead of its arXiv link, with the printed byline order confirmed on the publisher's pages; and (ii) a transposed article number in the Su, Guo and Wang (2024) entry was corrected (Information Sciences 676, article 120799, DOI 10.1016/j.ins.2024.120799 — the previously cited 120783 resolves to an unrelated article). One preprint is retained and explicitly flagged: Zhao et al. (arXiv:2604.07639) is the framework this paper studies; we verified via the arXiv record, the authors' official listing, and a Crossref registry query that no peer-reviewed version exists to date, the entry is labelled "Preprint", and every invocation of its theorems is co-cited with peer-reviewed anchors (Kallaugher, FOCS 2021; Huang et al., Science 2022; Kallaugher et al., SOSA 2025). A citation–bibliography consistency check confirms every in-text key maps one-to-one onto the bibliography. Following the Academic Editor's request that preprints not be cited, the remaining preprint's in-text citations were further reduced to the attribution-essential minimum, an explicit status footnote was added at its first citation in the Introduction, and one further peer-reviewed anchor was added and co-cited — Kallaugher, Parekh and Voronova (STOC 2024, DOI 10.1145/3618260.3649709), which establishes an exponential quantum–classical space separation for a natural streaming problem (this also corrects a Related-Work sentence that previously credited the 2021 FOCS paper with an exponential separation, whereas the advantage established there is polynomial). The bibliography now has 55 entries, consistency re-verified $55 \leftrightarrow 55$; a separate response to the Academic Editor accompanies this letter.

A tracked-changes PDF (`tracked_changes.pdf`, latexdiff against the reviewed version) accompanies this letter.

Point-by-point response

We thank the reviewer for an unusually thorough review; the closing recommendation — make the (τ, r) landscape-as-hypothesis-generator the paper's main message — is precisely the repositioning this revision performs.

R2.1 — "The key issue is the relationship between τ and r and the quantities in the framework of quantum oracle-sketching. … They have to strengthen the theoretical basis for the chosen proxy construction and distinguish the results from the original from the assumptions that are used in this paper."

Response. New §3.3 does this separation explicitly, in four labelled parts: *Established theory* (the $\times R$ overhead theorem, $O(n)$-qubit memory, task-specific O(N^c) hardness — all results of the original framework, taken as given); *Assumptions* (two-scalar summarisability, binarised-stream fidelity, task-rate-as-pipeline-stand-in — all modelling choices of this paper, none inherited); *Heuristic constructions* (the τ division absent from the framework's accounting; the VC-type rate in place of oracle-construction cost); and *Scope limitations*. Table 1 gives per-quantity provenance with an explicit "not guaranteed" column. We also now state the sharpest consequence: under the framework's own r-only accounting, the Markov, seasonal, and long-memory regimes ($r = 1$) incur zero overhead and no proxy–baseline crossover exists — the τ-penalised landscape is a classical effective-sample-size stress test, strictly more pessimistic than the framework's accounting, not a rendering of it. *Changes:* New §3.3 + Table 1; §3.2; effective-sample-size lineage now cited

(Geyer 1992).

R2.2 — "The terminology of quantum performance should be moderated throughout the paper. … we will refer to them as proxy-based predictions instead of the actual quantum performance. This is particularly relevant in the Abstract, Results, Discussion, figure captions and Conclusion."

Response. A systematic terminology pass was made over exactly those surfaces. "Quantum-favourable" no longer appears (now "proxy-favourable"/"hypothesized favourable"); "quantum performance proxy crosses below classical performance" and similar constructions were rewritten ("the accuracy proxy computed with the analytic τ lies below the measured classical baseline"); Table 6's column is "Proxy hypothesis"; and — because captions alone cannot fix text baked into images — Figures 1, 2, and 6 were regenerated to remove the in-image labels "n = 10 qubits", "Quantum proxy (IID, 10 qubits)", and "Quantum O(n)". Explicit "no quantum algorithm is implemented" statements now appear in the Abstract, §3.3, §6.5, §6.6, §6.8, §7.4, §8, and the Conclusion. *Changes:* Global; Figures 1, 2, 6 regenerated; all captions audited.

R2.3 — "The statistical comparison between classical baselines and the proxy model is still not resolved. Our Wilcoxon tests compare stochastic classical results for seeds to a proxy value that is deterministic for a given value of parameters. … We believe the magnitude of the classical minus proxy difference and the uncertainty may be a better gauge than simply p-values."

Response. Adopted in full — magnitude and uncertainty now lead, and the tests are re-scoped to the one thing they can support. Table 4 reports, per sweep point, the Hodges–Lehmann estimate of the paired difference with its exact signed-rank 95% interval (e.g., +0.042 [+0.011, +0.087] at $\rho = 0$; −0.046 [−0.079, −0.003] at $\rho = 0.88$) and the rank-biserial correlation, which we interpret explicitly as sign-concordance across seeds (scale-free against a near-constant reference; all magnitude information lives in the difference estimate). The p-values are descriptive companions only; we disclose that none survives Holm correction at $n = 10$ and why; we disclose the integer-τ lattice sensitivity (one lag step = 0.012, enough to move the $\rho = 0.48$ interval across zero); and the interpretation paragraph states that the crossover is the deterministic solution of $\text{acc\_proxy}(\hat{\tau}(\rho))$ = mean classical accuracy, with uncertainty inherited from the estimator and the constants (C, δ), not from the tests. The full re-analysis (which first reproduced the published per-seed results bit-exactly — all five p-values to six digits) ships with the revision. *Changes:* Table 4 + caption; §6.5 (rewritten body and interpretation paragraphs); Abstract; §7.1; §8; `verification_scripts/exp5_reanalysis.py` + output.

R2.4 — "The huge discrepancy between the analytic and empirical refreshing time is one of the most important findings in the paper, especially for the long-range dependence regime. … The authors should discuss this in more detail and discuss its implications for the interpretation of the central (τ,r) sensitivity landscape."

Response. We agree it is one of the paper's most important findings and have promoted it accordingly. New §7.3 gives the mechanism (the analytic proxies target model-level worst-case scales — joint-chain mixing; tail-dominated integrated autocorrelation — while the empirical estimator measures the short-lag marginal decorrelation of the binarised stream; for a non-mixing process there is no scalar the two could agree on), and the implications for the landscape are now carried through the whole paper: verdicts are bound to their estimator convention everywhere, and we state openly that the burst and long-memory extremes *exchange places* between the analytic and empirical conventions (burst saturates through its measured duplication; long-memory's high empirical value is the short-lag artefact) — in Experiment 9, §7.3, the RQ2/RQ3 answers, and the Figure 3/7 captions. The abstract now says the two estimates "diverge" (not "overestimate", which presupposed the empirical value as ground truth) and adds that they answer different questions. *Changes:* New §7.3; §6.9; §6.3; Figures 3, 7 captions; Conclusion RQ1–RQ3; Abstract.

R2.5 — "The classical-memory comparison needs to be taken with care. The classical lower bound curve … is not actually an empirical lower bound for the SGD, averaged-SGD, or Count-Min tools used in our experiments. … we should make it more prominent in the dimension-scaling discussion."

Response. Made prominent exactly there. Experiment 6 now opens with: "No memory lower bound is measured empirically in this experiment: the curves in panel (a) plot cited theoretical quantities, and no result here constitutes an empirical memory floor for online SGD, averaged SGD, or Count-Min." Figure 6's caption labels panel (a) "cited bounds; no measured content" and the in-image curve labels now read "(cited theory)". §3.3's "Memory references" paragraph and the provenance table carry the same statement structurally, and §7.1 states that none of its structural factors is a measured quantum result. *Changes:* §6.6; Figure 6 (caption and regenerated labels); §3.3; Table 1; §7.1.

R2.6 — "The real-stream analysis is useful as a sanity check but it is limited to two NAB datasets. The conclusions on the sensor and telemetry applications should therefore be conservative."

Response. The conclusions are now explicitly conservative: §6.8 states the experiment "is illustrative rather than statistically representative", the External-validity paragraph records that generalisation is unverified, and the abstract claims placement only. Additionally, a new encoder/resolution sensitivity table strengthens what *can* be said from two streams: across ten re-encodings of the same raw series (three sampling resolutions × three bucket widths), the NYC-above-Machine-Temperature ordering holds in every configuration, so the reported placement is at least encoder-stable. Broader corpora are committed in Future Work with a concrete test protocol (H1). *Changes:* §6.8; new encoder-sensitivity paragraph + table; §8; §9; §7.4 H1.

R2.7 — "the target defined based on a percentile of the full time series should also be reconsidered in the context of streaming analysis, since using the full series to set the target threshold can take into account future observations. A threshold set from a first calibration interval or updated only from previously observed samples would provide a more rigorous prequential design."

Response. This is an excellent catch, and we now disclose it explicitly rather than leave it implicit. §6.8 ("Binarisation choice") states that the threshold is computed from the full series and therefore uses information not available prequentially at each step; that the feature stream and the (τ, r) estimates involve no such look-ahead — so the landscape placement, the paper's central output for these streams, is unaffected, the leakage touching only the classical-baseline reference numbers; and that a threshold calibrated on an initial interval, or updated from previously observed samples only, is the rigorous prequential design, which we adopt as future work. The Threats section (Internal validity, item (v)) records it as a residual risk. We chose disclosure over re-running the baselines in this revision to keep every published number bit-reproducible; we are happy to re-run with a calibration-interval threshold in a further round if the reviewer prefers. *Changes:* §6.8 Binarisation paragraph; §8 Internal validity (v); §9.

R2.8 — "The interpretation of the NYC Taxi Count-Min result also needs to be qualified. The six-bit representation gives a relatively small number of possible feature vectors and hence memorising the same feature patterns is relatively easy. … I would like to give more weight to the implications when we think about the apparent agreement between the Count-Min performance and our proxy prediction."

Response. The qualification now carries real weight in three places. The dedicated "Why Count-Min looks strong" paragraph is retained; the interpretation adds, plainly: "the favourable-side separation between the two streams rests on Count-Min alone; the hashed-linear learners do not corroborate it" — and we additionally report that both SGD variants fall *below* the majority baseline on accuracy and that Averaged-SGD's balanced accuracy straddles chance, so the apparent proxy–Count-Min agreement is explicitly not read as validation. §7.4 lists representation-specific effects like this memorisation route among the things the landscape *cannot* anticipate, and — going further in the reviewer's direction — the same section now concedes that, read strictly as an accuracy projection, the proxy is falsified on Machine Temperature (0.500 predicted vs 0.72–0.80 observed), with the rare-class diagnostic reading offered as interpretation, not confirmation. Table 6 now carries the ordering-to-ordering intervals (Count-Min F1 = 0.644 ± 0.062, per-ordering range 0.43–0.72), which tempers the headline number. *Changes:* §6.8 (findings bullets, Count-Min paragraph, interpretation (ii)–(iii)); Table 6 (intervals restored); §7.4.

R2.9 — "In particular, the target function ablation should be highlighted because it shows that our classical performance can change dramatically with the classification task in question and that our proxy is in fact target independent. A correlation-based proxy that is insensitive to the target function cannot necessarily predict the performance of algorithms for different kinds of decision problems."

Response. Highlighted and stated as a scope boundary in the reviewer's own terms. §6.9 (Target function) now reads: "classical learner performance is strongly target-dependent, while the proxy depends only on (τ, r, T) and is target-independent by construction. The proxy therefore cannot universally predict downstream classifier performance; it characterises the correlation structure of the stream — the effective sample budget available to any bounded-memory learner — not task-specific learnability." §7.4 elevates this to the "cannot" list (with the further empirical point that measured classical accuracy is flat while T_eff falls fivefold, so the budget model does not predict even the classical arm's accuracy), the Figure 9 caption states the proxy "does not track task-specific learnability", and RQ2's answer now says τ and r are budget summaries, not accuracy predictors. The ablation's protocol (prequential, lr = 0.01, T = 4096) is also now stated so its absolute levels are not misread against the main experiments. *Changes:* §6.9; §7.4; Figure 9 caption; Conclusion RQ2.

R2.10 — "the strongest contribution of the paper is the (τ,r) sensitivity landscape proposed as a hypothesis-generating tool … The paper would be better off having this methodological contribution as the main message rather than the separation of quantum-classical performance."

Response. This is now the paper's stated identity, end to end: the title leads with "Proxy-Based Diagnostics"; the abstract announces "a proxy-based diagnostic and hypothesis-generation framework"; the contribution list names the landscape "the paper's central diagnostic and hypothesis-generation artifact"; §7.4 states the falsifiable hypotheses it generates; and the Conclusion opens by declaring the landscape the primary contribution and hypothesis generation the secondary one, followed by: "The paper demonstrates no quantum advantage and implements no quantum algorithm." *Changes:* Title; Abstract; §1; §7.4; §10.

Note on verification

Before making any statistical change we reproduced the published Experiment-5 per-seed results bit-exactly from the pinned seeds (all five Wilcoxon p-values match the published values to six significant digits), and all figure regenerations are rendering-only (numerical outputs byte-identical). The re-analysis and sensitivity scripts, with their outputs, are included in the accompanying package (`verification_scripts/exp5_reanalysis.py` and `verification_scripts/encoder_sensitivity.py`; `code/exp5_effectsizes.py` in the public repository), and the reproducibility documentation was corrected and extended (exact seed indices per experiment; the previously missing Machine-Temperature download step).

On behalf of all authors,

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